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LEARNER GUIDE

For

Effective Project Management for Small Projects

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17.0	19 August 2026	Legacy master trainer slide release used as the controlling syllabus.	Tertiary Infotech Academy
18.0	19 August 2026	Complete visual and instructional redesign; content enriched from the Agile and PMP references; seven connected tool-based activities and self-contained editable template packs added; LG and LP regenerated.	Dr Alfred Ang

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1. Course Overview

This guide accompanies the visual trainer deck. Detailed procedures live here and in the self-contained activity folders; the PowerPoint intentionally contains only activity briefings and success criteria.

1.1 Learning Outcomes

- LO1: Develop project scope and team to initiate the facilitation of a small project
- LO2: Develop a realistic project plan based on the agreed project scope
- LO3: Utilise appropriate methods and tools to track and drive a project
- LO4: Identify, analyse and manage project risks
- LO5: Collaborate effectively with project stakeholders
- LO6: Deploy manpower, budget and other resources for efficient completion
- LO7: Track project deliverables and correct misalignments through control and closeout

1.2 Skills Framework Alignment

Code	Ability / Knowledge	Evidence
A1	Facilitate execution of small projects that are typically less than six months, with limited budget and interdependencies, and no significant strategic impact	Activity 1
A2	Implement realistic project plans based on project objectives and scope	Activity 2
A3	Utilise appropriate methods and tools to track and drive progress against plans and timelines	Activity 3
A4	Identify risks to project success and take appropriate actions to manage them	Activity 4
A5	Collaborate effectively with relevant internal and external stakeholders directly impacting the project	Activity 5
A6	Deploy manpower, financial budgets and relevant resources for efficient and effective completion	Activity 6
A7	Track deliverables against schedules, monitor costs,	Activity 7

	timescales and resources, and take basic corrective actions when misaligned	
K1	Elements of a small project	Topic 1
K2	Requirements of a project plan	Topic 2
K3	Application of appropriate project management methodologies and tools	Topic 3
K4	Project risks	Topic 4
K5	Project stakeholder identification	Topic 5

1.3 Running Case

A 42-person Singapore specialty grocer will introduce online ordering and two-hour click-and-collect at three outlets. The five-month pilot has a S\$180,000 cap, a core team of seven, two external suppliers, and limited dependencies. It is important to operations but does not change the company's overall strategy, making it a genuine small project.

Success target: Launch at all three outlets by 30 January 2027; process at least 120 pilot orders per week; keep pick accuracy at or above 98%; achieve customer satisfaction of at least 4.2/5; and remain within the approved S\$180,000 budget.

1.4 Tool-Based Evidence

Each activity folder includes the relevant editable Office tools from the source template library. Save completed tools under the activity folder or your LMS submission workspace with clear versioned filenames.

Topic 01 — Initiation of a Project

Alignment: A1, K1 · LO1

Small-project fit · value · charter · scope · team · kickoff

Why This Topic Matters

Use the small-project screen before tailoring governance. A five-month, S\$180,000 pilot with seven people needs proportionate controls, but it still needs authorisation, baselines and decisions.

Core Concepts

- A project is temporary, unique and value-seeking; operations are ongoing and repetitive.
- A small project is normally under six months, has a limited budget and team, few dependencies and no significant strategic impact.
- The project manager integrates people, decisions and trade-offs; they do not merely maintain a schedule.
- Success balances value and outcomes with scope, schedule, cost, quality, resources and risk.
- Predictive, adaptive and hybrid approaches are choices driven by uncertainty, not labels of maturity.
- The business case tests whether the project should exist; the charter authorises it to exist.
- A useful charter names purpose, measurable success, scope boundaries, milestones, budget, major risks, team and sponsor authority.
- A kickoff creates one shared picture of why, what, who, how and what happens next.

Applied Guidance

Use the small-project screen before tailoring governance. A five-month, S\$180,000 pilot with seven people needs proportionate controls, but it still needs authorisation, baselines and decisions.

Business value includes financial return, service improvement, risk reduction and capability. Do not combine these into one unsupported number.

A charter is strongest when it exposes what the project will not do and what the project manager may decide without returning to the sponsor.

For hybrid work, keep the charter and milestone gates while allowing short build-and-feedback cycles inside the delivery phase.

Activity 1 — Authorise the HarbourHub Click & Collect Pilot

Field	Detail
Duration	75 minutes
Objective	Develop project scope and team to initiate

	a five-month small project (A1, K1).
Tools	Project Charter, Cost–Benefit Analysis and Business Case templates
Situation	HarbourHub Fresh has approved discovery work for a click-and-collect pilot at three outlets, but the sponsor has not yet authorised delivery. Operations wants speed, Finance wants a hard S\$180,000 ceiling, store managers fear disruption, and customers want reliable collection times. Your team must decide whether this is a viable small project and define the authority to proceed.
Output	A cost–benefit analysis, one-page business case and signed-ready project charter with measurable success criteria, scope boundaries, milestone dates, budget cap, team and kickoff agenda.

Tool Files

- 02 Cost Benefit Analysis.xlsx
- 05 Business Case on a Page.docx
- 08 Project Charter Template.xlsx

Detailed Procedure

1. Read the HarbourHub case and test it against the small-project criteria: temporary, unique, under six months, limited budget, limited interdependencies and no significant strategic impact.
2. Open the Design Thinking tool and capture the primary user: an outlet supervisor balancing walk-in customers, online orders and collection queues.
3. Write one evidence-based problem statement in the form: user + need + consequence. Do not embed a solution in the need.
4. Open 02 Cost Benefit Analysis.xlsx. Enter the S\$180,000 project cap, anticipated first-year benefits, transition costs and assumptions. Separate cash benefits from non-financial benefits.
5. Calculate net benefit, payback period and the sensitivity of the case to 20% lower order volume. Record which assumption is most decision-sensitive.
6. Complete 05 Business Case on a Page.docx: problem, options, recommended option, value, major risks and decision required.
7. Open 08 Project Charter Template.xlsx and define purpose, SMART success measures, in-scope deliverables and explicit exclusions.

8. Record the five-month milestone frame, S\$180,000 cap, major assumptions, constraints and the top three high-level risks.
9. Name one sponsor, one project manager and the seven-person core team. State the project manager's decision authority and escalation threshold.
10. Prepare a 15-minute kickoff agenda covering why, what, who, how, first decisions and immediate actions.
11. Run a sponsor gate review with another team. They may approve, approve with conditions or reject. Record the decision and conditions.

Acceptance Criteria

The charter proves the work qualifies as a small project, names one accountable sponsor, contains quantified success measures and exclusions, stays within five months and S\$180,000, and carries an explicit authorisation decision.

Debrief

- The business case answers whether the project should exist; the charter answers whether the team is authorised to act.
- A scope boundary is incomplete until exclusions are visible—silence is where scope creep starts.
- Success criteria must be measurable before the plan is built, or control later becomes opinion.

Topic 02 — Develop a Project Plan

Alignment: A2, K2 · LO2

Requirements · scope baseline · WBS · network · schedule · budget

Why This Topic Matters

Start requirements with the stakeholder need, then write acceptance evidence. Words such as easy, fast and user-friendly are not testable until a measure or observation is defined.

Core Concepts

- The plan integrates subsidiary decisions; separate scope, schedule and cost files are not yet a project plan.
- Progressive elaboration and rolling-wave planning add detail as uncertainty reduces without abandoning control.
- Requirements must be testable, prioritised and traceable from need through deliverable and acceptance evidence.
- The scope statement defines deliverables, exclusions, constraints, assumptions and acceptance criteria.
- The WBS decomposes deliverables to manageable work packages; the WBS dictionary removes ambiguity.
- Dependencies create the network; the longest path sets the earliest finish and carries zero or least float.
- Duration estimates should separate work, waiting, risk and resource availability; use analogous, parametric, bottom-up or three-point methods deliberately.
- The cost baseline is time-phased; contingency reserve covers known-unknowns while management reserve covers unknown-unknowns.

Applied Guidance

Start requirements with the stakeholder need, then write acceptance evidence. Words such as easy, fast and user-friendly are not testable until a measure or observation is defined.

The WBS follows deliverables rather than the organisation chart. Work packages should be large enough to manage and small enough to estimate, assign and accept.

The critical path is the longest-duration path through the current network. It is not necessarily the most technically difficult or politically important work.

A baseline is the approved reference for control. A forecast is the latest expected outcome. Keep both visible; overwriting the baseline destroys the ability to explain variance.

Activity 2 — Build the Integrated Scope, Schedule and Cost Baseline

Field	Detail
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Duration	120 minutes
Objective	Implement a realistic project plan from the authorised objectives and scope (A2, K2).
Tools	RTM, WBS, network, Gantt and budget templates
Situation	The sponsor approved the charter with one condition: the three outlets must launch before Lunar New Year trading. You have high-level deliverables but no traceable requirements, work breakdown, dependency logic or time-phased budget. A vendor has proposed a date based only on optimism.
Output	A requirements traceability matrix, WBS and dictionary, network diagram with critical path and float, Gantt schedule with milestones, and time-phased S\$180,000 cost baseline with contingency reserve.

Tool Files

- 24 WBS Dictionary.xlsx
- 24 Work Breakdown Structure.pptx
- 25 Requirements Traceability Matrix.xlsx
- 28 Schedule Network Diagram.xlsx
- 29 - Gantt Chart - Schedule - Daily Weekly Monthly and Phases.xlsx
- 32 Project Budget - Planned vs Actual - Weekly and Monthly.xlsx

Detailed Procedure

1. Review Activity 1's charter. Copy every success measure, constraint and in-scope deliverable into the planning assumptions log.
2. Open 25 Requirements Traceability Matrix.xlsx and write at least eight testable requirements. Give each an ID, source, priority, deliverable, acceptance evidence and owner.
3. Apply MoSCoW. Challenge every Must: if removing it does not prevent the pilot from achieving its success criteria, it is not a Must.
4. Open 24 Work Breakdown Structure.pptx. Decompose by deliverable to work packages that can be estimated, assigned and accepted. Apply the 100% rule.
5. Complete 24 WBS Dictionary.xlsx for at least eight work packages: description, owner, acceptance criteria, assumptions, dependencies, estimate and milestone.
6. Enter the supplied activity list in 28 Schedule Network Diagram.xlsx. Add predecessor logic and identify all finish-to-start, start-to-start and lag relationships.

7. Perform the forward pass and backward pass. Mark the critical path and calculate total float for every non-critical activity.
8. Open the Gantt template and enter the work packages, owners, durations, dependencies and three decision milestones. Keep the project within five months.
9. Estimate cost using at least two methods—bottom-up for work packages and analogous or parametric for a cross-check. Record the basis of estimate.
10. Open the budget template. Time-phase labour, vendor, equipment, training and contingency. Keep the cost baseline plus contingency within S\$180,000.
11. Run an integration check: every requirement traces to a WBS deliverable; every work package appears in the schedule and budget; every milestone has acceptance evidence.
12. Ask another team to find one orphan requirement, missing dependency or unfunded work package. Repair the baseline and record the change.

Acceptance Criteria

Every requirement traces to acceptance evidence and a WBS deliverable; the network produces a defensible critical path; the Gantt fits the five-month limit; and all scheduled work is funded within S\$180,000.

Debrief

- A Gantt chart is an output of dependency logic, not a substitute for it.
- The integrated baseline is only credible when scope, schedule and cost reconcile to the same work packages.
- Float belongs to the project, not automatically to the activity owner who happens to see it.

Topic 03 — Apply Methods and Tools to Drive a Project

Alignment: A3, K3 · LO3

Quality planning · flow · root cause · Pareto · improvement

Why This Topic Matters

Tool choice begins with the question. Use a check sheet for facts, fishbone for breadth, 5 Whys for depth, Pareto for priority and a control chart for behaviour over time.

Core Concepts

- A method defines how work is managed; a tool makes information visible. Choose both for the decision needed.
- Predictive plans optimise sequence and baseline control; adaptive boards optimise flow and feedback; hybrid combines both intentionally.
- Quality means conformance to requirements and fitness for use, not luxury or perfection.
- Prevention and appraisal are usually cheaper than internal and external failure.
- A check sheet captures facts, a control chart detects process behaviour, Pareto prioritises causes, and fishbone plus 5 Whys investigates root cause.
- Root-cause analysis must end at a process or system cause the team can influence, not blame a person.
- Visual work management exposes blocked work and excessive WIP before the schedule report does.
- Continuous improvement closes the loop through a named action, owner, measure and review date.

Applied Guidance

Tool choice begins with the question. Use a check sheet for facts, fishbone for breadth, 5 Whys for depth, Pareto for priority and a control chart for behaviour over time.

Common-cause variation belongs to the process; special-cause variation has an identifiable event or condition. Tampering with a stable process after every data point can make performance worse.

Corrective action addresses an observed variance or non-conformity. Preventive action lowers the probability of future variance. Both need an owner and verification.

Visual work management should expose blocked work, WIP, ageing and ownership—not become a colourful task list with no decision rules.

Activity 3 — Diagnose and Prioritise the Pilot's Quality Failure

Field	Detail
Duration	75 minutes
Objective	Use appropriate methods and tools to drive progress and improve quality (A3,

	K3).
Tools	Fishbone, Check Sheet, 5 Whys, Pareto and Control Chart
Situation	During outlet testing, only 82% of orders are ready at the promised collection time and pick accuracy has fallen to 94%. The software vendor blames store staff; store staff blame late inventory updates. The sponsor wants action before the team has separated symptoms, causes and normal variation.
Output	A reconciled check sheet, fishbone diagram, five-level causal chain, Pareto chart, control-chart interpretation and a SMART corrective experiment with owner and review date.

Tool Files

- 35 Check Sheets.xlsx
- 37 Ishikawa Diagram.xlsx
- 38 Five Whys.xlsx
- 36 Pareto Principle Template.xlsx
- 34 Control Chart.xlsx

Detailed Procedure

1. Open 35 Check Sheets.xlsx and enter the supplied 60-order test dataset by defect type, outlet, shift and date.
2. Reconcile the total. Separate late-ready orders from wrong-item, stock-out, notification and payment defects.
3. Open the Fishbone browser tool or 37 Ishikawa Diagram.xlsx. Set the effect to 'Orders not ready accurately at the promised time'.
4. Generate causes across People, Process, Technology, Data, Measurement and Environment. Mark facts with F and hypotheses with H.
5. Select the highest-frequency cause family and complete 38 Five Whys.xlsx until the root is a process or system cause the team can influence.
6. Read the chain backwards with 'because'. Repair any causal jump or circular statement.
7. Open 36 Pareto Principle Template.xlsx and rank the defect categories. Add the cumulative percentage and identify the crossing of 80%.
8. Use 34 Control Chart.xlsx to plot daily ready-on-time percentage. Decide whether the signal suggests a stable process or special-cause behaviour.

9. Propose one SMART improvement experiment with an owner, target, completion date and measure. Allocate it on the project Kanban board.

10. Explain to the sponsor why fixing every listed cause at once would be slower and less controlled than testing the vital few.

Acceptance Criteria

The data total reconciles; Pareto reaches 100%; the five-whys chain ends at an actionable cause; the control chart interpretation matches the pattern; and the corrective experiment has an owner, measure and review date.

Debrief

- Fishbone broadens the search, 5 Whys deepens one chain, and Pareto tells you which chain is worth funding first.
- A control limit describes process behaviour; it is not the same as a customer specification limit.
- Improvement becomes project work only when it has capacity, ownership and a verification date.

Topic 04 — Manage Project Risk

Alignment: A4, K4 · LO4

Identify · assess · respond · contingency · monitor

Why This Topic Matters

A risk statement separates cause, uncertain event and impact. This makes the response design testable: act on the cause, reduce probability, reduce impact or prepare contingency.

Core Concepts

- Risk is an uncertain event or condition with a positive or negative effect; an issue has already happened.
- Write risks as cause-event-impact statements so the response targets the cause or reduces the impact.
- The risk register records owner, probability, impact, exposure, triggers, response, contingency and residual risk.
- Qualitative analysis prioritises quickly; quantitative analysis estimates the range of outcomes and decision value.
- Threat responses are avoid, mitigate, transfer, accept or escalate; opportunity responses are exploit, enhance, share, accept or escalate.
- A contingency plan is triggered by an observable threshold and funded from contingency reserve.
- Decision trees compare expected monetary value; tornado charts reveal which assumptions deserve the most attention.
- Risk reviews are recurring decisions, not a register created once and archived.

Applied Guidance

A risk statement separates cause, uncertain event and impact. This makes the response design testable: act on the cause, reduce probability, reduce impact or prepare contingency.

Probability-impact scores are ordinal prioritisation aids. Do not pretend that multiplying two subjective ratings creates a precise financial forecast.

Expected monetary value helps compare repeated or material choices. It can still miss reputation, safety, legal duty and strategic option value.

Residual risk remains after the response; secondary risk is introduced by the response. Both return to the register with owners.

Activity 4 — Build the Risk Register and Fund the Right Responses

Field	Detail
Duration	75 minutes
Objective	Identify risks to project success and take

	proportionate action (A4, K4).
Tools	Risk Matrix, Decision Tree and Tornado templates
Situation	The pilot has entered build. The POS interface may be delivered late, refrigerated stock counts are unreliable at one outlet, a key trainer is available for only two weeks, and a faster payment option could improve adoption but costs more. The remaining contingency reserve is S\$18,000.
Output	A prioritised risk register with owners, triggers, responses, contingency and residual exposure; one decision-tree recommendation; one tornado sensitivity reading; and a reconciled S\$18,000 reserve plan.

Tool Files

- 48 Risk Assessment Matrix Two pages.xlsx
- 45 Decision Tree Analysis TDIDT.xlsx
- 49 Tornado Chart - Sensitivity Analysis.xlsx

Detailed Procedure

1. Open the risk matrix template and enter at least eight risks using the form: because of cause, event may occur, leading to impact.
2. Separate risks from current issues. Move anything already occurring into the issue log and give it an action owner.
3. Score probability and impact using the template scale. Calculate exposure and justify the top three priorities.
4. Assign one risk owner and one response owner. Define an observable trigger for each top risk.
5. Choose avoid, mitigate, transfer, accept or escalate for threats; use exploit, enhance, share, accept or escalate for opportunities.
6. Specify preventive action, contingency action and fallback for the POS-interface risk. State when each activates.
7. Open the decision-tree template. Compare the standard payment option with the faster premium option using the supplied probabilities, costs and benefits.
8. Calculate expected monetary value and recommend an option. Then state one qualitative factor EMV does not capture.

9. Open the tornado template and vary adoption, order volume, vendor delay and training productivity. Identify the two most sensitive assumptions.
10. Allocate the S\$18,000 contingency reserve to named risks. Do not consume management reserve or hide general padding inside activities.
11. Conduct a risk review: close obsolete risks, update exposure, promote triggered risks to issues and record new secondary risks created by responses.

Acceptance Criteria

Every priority risk has cause-event-impact wording, an owner, trigger, funded response and residual score; the decision-tree arithmetic reconciles; and the named contingency allocations total no more than S\$18,000.

Debrief

- An owner does not prevent a risk; they watch the trigger and ensure the agreed response is ready.
- Contingency is not generic padding—it is a funded response to identified uncertainty.
- Quantitative tools sharpen a decision but do not replace judgement about reputation, safety or compliance.

Topic 05 — Collaborate with Stakeholders

Alignment: A5, K5 · LO5

Identify · analyse · communicate · engage · allocate accountability

Why This Topic Matters

Stakeholder identification is continuous. Procurement, design choices, issues and change requests can create new stakeholders or alter influence and urgency.

Core Concepts

- A stakeholder can affect, be affected by, or perceive themselves to be affected by the project.
- The stakeholder register captures interest, influence, expectations, impact, communication needs and engagement strategy.
- Power-interest mapping guides attention; the salience model adds legitimacy and urgency.
- The engagement assessment matrix compares current and desired states so the gap becomes actionable.
- RACI clarifies who is Responsible, Accountable, Consulted and Informed; every deliverable needs one clear Accountable owner.
- Communication planning matches audience, message, channel, frequency, owner and feedback route.
- Engagement is two-way sense-making and decision participation, not one-way broadcasting.
- Escalation should state the decision needed, options, impact of delay and recommendation.

Applied Guidance

Stakeholder identification is continuous. Procurement, design choices, issues and change requests can create new stakeholders or alter influence and urgency.

Power-interest mapping guides management attention; salience adds legitimacy and urgency. Use both when formal authority does not explain actual influence.

A RACI matrix is a diagnostic for decision rights. One Accountable owner is the default; exceptions need an explicit governance reason.

A communication plan specifies message, audience, channel, frequency and owner. An engagement plan adds the feedback route and decision the stakeholder can shape.

Activity 5 — Align Stakeholders with a Map, RACI and Engagement Plan

Field	Detail
Duration	75 minutes
Objective	Collaborate with internal and external stakeholders directly impacting the project

	(A5, K5).
Tools	Stakeholder Map, Saliency, RACI and Engagement Assessment
Situation	Two weeks before outlet testing, the POS vendor and operations manager each believe the other owns interface acceptance. Store managers receive conflicting instructions, Finance has not approved the payment milestone, and a high-influence outlet supervisor is quietly resisting the pilot.
Output	A stakeholder register and power-interest map, saliency reading, validated RACI matrix, current-to-desired engagement assessment, communication plan and one evidence-based escalation brief.

Tool Files

- 42 Stakeholder Impact Over Influence Matrix.xlsx
- 43 Saliency Chart.pptx
- 39 RACI Template.xlsx
- 41 Stakeholder Engagement Assessment Matrix.xlsx

Detailed Procedure

1. List at least 12 internal and external stakeholders: sponsor, PM, store managers, outlet staff, customers, Finance, IT, POS vendor, payment vendor, trainer and support team.
2. Open the impact-over-influence template and position every stakeholder. Write the evidence for each high-influence rating.
3. Use the saliency template to test power, legitimacy and urgency. Compare the result with the power-interest map and explain one difference.
4. Open the RACI browser tool or 39 RACI Template.xlsx. Use rows for deliverables and decisions, not job titles or vague activities.
5. Assign R, A, C and I. Enforce one accountable owner per row; flag rows with no R, no A, too many Cs or duplicated As.
6. Resolve interface acceptance, payment release, training sign-off, launch approval and change control. Record why each A sits where it does.
7. Open the engagement assessment matrix and rate current versus desired engagement for the top eight stakeholders.
8. For each material gap, define message, channel, frequency, owner, feedback mechanism and the next decision the engagement supports.

9. Draft a one-minute escalation brief for the interface ownership conflict: decision required, options, impact of delay, recommendation and deadline.
10. Role-play the meeting with another team acting as sponsor and vendor. Update the RACI and stakeholder plan from the feedback.

Acceptance Criteria

All material stakeholders are analysed; every critical deliverable has exactly one Accountable owner and at least one Responsible party; engagement gaps have two-way actions; and the escalation asks for a clear decision.

Debrief

- Most collaboration failures are decision-right failures before they become communication failures.
- RACI clarifies accountability; it does not replace the conversation needed to gain commitment.
- Stakeholder attention should follow evidence of influence, impact and urgency—not organisational seniority alone.

Topic 06 — Deploy Resources

Alignment: A6 · LO6

People · budget · equipment · procurement · team performance

Why This Topic Matters

Resource availability is the fraction of time usable for this project after operations, leave and other commitments—not the employee's full calendar.

Core Concepts

- Resource planning links each work package to required skills, quantities, timing and availability.
- A responsibility assignment is not a capacity plan; availability must be tested against concurrent work.
- Resource levelling changes dates to resolve over-allocation; smoothing stays within available float.
- Fast tracking overlaps work and raises risk; crashing adds cost to critical-path work and may not shorten the project.
- Procurement decisions consider make-or-buy, lead time, quality, interfaces, contract type and vendor capability.
- Tuckman's stages require adaptive leadership: direct, coach, support and delegate as the team matures.
- Conflict is normal; collaborate for important relationships and outcomes, compromise when time is limited, and escalate only with evidence.
- A resource dashboard should show demand, capacity, variance, bottlenecks and decisions—not just headcount.

Applied Guidance

Resource availability is the fraction of time usable for this project after operations, leave and other commitments—not the employee's full calendar.

Smoothing uses available float and protects the end date. Levelling resolves over-allocation even when the project finish must move.

Crashing adds cost to critical-path work; fast tracking overlaps work that was sequential. Recalculate the network and risk after either action.

Conflict mode should match outcome importance and relationship importance. Collaboration is powerful but costly; not every low-stakes disagreement needs it.

Activity 6 — Deploy the Team and Resolve the Resource Bottleneck

Field	Detail
Duration	60 minutes
Objective	Deploy manpower, budget and relevant resources for efficient completion (A6).

Tools	Organisation Chart, Kanban Board and Project Budget templates
Situation	The test plan needs 22 person-days in the next two weeks, but only 16 are available. The same business analyst is assigned to testing, training materials and vendor clarification. The sponsor refuses a launch delay unless the team proves no lower-cost option exists.
Output	A two-week capacity plan, organisation and escalation map, workload Kanban, evaluated compression options, reconciled budget impact and a sponsor recommendation that removes the bottleneck.

Tool Files

- 40 Organisation Chart.pptx
- 30 Kanban Board.xlsx
- 32 Project Budget - Planned vs Actual - Weekly and Monthly.xlsx

Detailed Procedure

1. Open the organisation-chart template and map the sponsor, PM, functional managers, vendor leads and seven-person core team. Show escalation lines separately from work coordination.
2. Create a two-week capacity table by person and skill. Use available person-days, not nominal headcount.
3. Map every near-term work package to skill, effort, owner, earliest start, dependency and due date.
4. Calculate demand minus capacity. Identify the business analyst as the constraint and quantify the six-person-day gap.
5. Open the Kanban template or Scrum browser tool. Create To Do, Ready, In Progress, Review and Done columns and set WIP limits around the analyst's constraint.
6. Test resource smoothing by using float on non-critical tasks. Recalculate the schedule and note whether the project finish changes.
7. Test resource levelling by moving work until demand fits capacity. Record the new finish date and milestone impact.
8. Test crashing by adding a contract analyst to critical-path work. Enter the incremental cost and check whether the saved time justifies it.
9. Test fast tracking by overlapping training-material drafting with late testing. Record the rework and quality risk introduced.

10. Update the project budget with the preferred option. Keep approved changes separate from contingency usage.
11. Recommend one option to the sponsor using four fields: schedule effect, cost effect, risk effect and decision deadline.
12. Define team-working rules for handoffs and conflict: who decides, how blockers are raised, and when the PM escalates.

Acceptance Criteria

The plan reconciles named work with actual capacity; the bottleneck is visible; each optimisation option has schedule, cost and risk consequences; and the chosen deployment fits the approved project constraints.

Debrief

- A name in a RACI matrix does not create capacity; capacity must be measured against concurrent demand.
- Fast tracking buys time with risk, while crashing buys time with cost—neither is automatically a good decision.
- Manage the constraint first. Adding capacity elsewhere can increase WIP without improving delivery.

Topic 07 — Track Project Deliverables and Correct Misalignments

Alignment: A7 · LO7

Performance · earned value · change · corrective action · handover · closure

Why This Topic Matters

PV, EV and AC must share the same status date and scope basis. Otherwise SPI and CPI compare unlike quantities.

Core Concepts

- Work performance data becomes information only after comparison with the approved baseline and thresholds.
- A status report should answer: where are we, why, what is forecast, what decision is needed, and who owns the action.
- Earned value integrates scope, schedule and cost: PV is planned work, EV is completed value, and AC is actual spend.
- $CPI = EV/AC$ and $SPI = EV/PV$; indices below 1.00 signal inefficiency against the baseline.
- A change request states reason, impact, options and recommendation; approved changes update baselines and traceability.
- Corrective action brings performance back to plan; preventive action reduces the probability of future variance.
- Schedule compression is applied to the current critical path and re-analysed after every change.
- Closure confirms acceptance, transition, contract completion, records, benefits ownership and lessons learned.

Applied Guidance

PV, EV and AC must share the same status date and scope basis. Otherwise SPI and CPI compare unlike quantities.

CPI and SPI below 1.00 are adverse. Translate indices into plain-language forecast and decision consequences rather than reporting decimals alone.

Change control is not change prevention. It makes trade-offs explicit, records authority and updates every affected baseline after approval.

Closure includes acceptance, transition, contracts, records, lessons and benefits ownership. The project may finish while benefits continue under operations.

Activity 7 — Control the Pilot, Decide the Change and Close the Project

Field	Detail
Duration	105 minutes
Objective	Track deliverables, monitor cost,

	schedule and resources, and correct misalignment (A7).
Tools	Kanban, Burndown, Budget, Control Chart and Retrospective templates
Situation	At the end of week 12, the baseline planned value is S\$108,000. Completed deliverables represent S\$90,000 of budgeted value and actual cost is S\$105,000. The sponsor also requests loyalty points, estimated at S\$24,000 and three weeks. Launch evidence is incomplete and the vendor wants final payment.
Output	A current Kanban and burndown, earned-value analysis with forecasts, status report, change-control decision, corrective-action log, acceptance and handover checklist, lessons learned and closure recommendation.

Tool Files

- 30 Kanban Board.xlsx
- 31 Burndown Chart.xlsx
- 32 Project Budget - Planned vs Actual - Weekly and Monthly.xlsx
- 34 Control Chart.xlsx
- 44 Agile Retrospective.xlsx

Detailed Procedure

1. Update the Kanban board with each deliverable, owner, status, blocker, due date and acceptance evidence. Do not count work as Done without evidence.
2. Open the burndown template and plot remaining work by week. Identify any flat period, late cliff or scope increase.
3. Record PV = S\$108,000, EV = S\$90,000 and AC = S\$105,000. Calculate SV, CV, SPI and CPI.
4. Interpret each result in plain language. State separately whether the project is ahead/behind schedule and under/over cost for work performed.
5. Using BAC = S\$162,000, calculate EAC under current cost performance and ETC. Compare the forecast with the S\$180,000 authorised ceiling.
6. Prepare a one-page status: traffic light with threshold, variance cause, forecast, top risk/issue, corrective action, owner and decision needed.
7. Document the loyalty-points request: reason, scope, schedule, cost, resource, quality and risk impact; options; recommendation; and decision deadline.

8. Apply the change-control workflow. Decide reject, defer to a later phase, or approve with trade-off. Record the approver and rationale.
9. If approved, update the RTM, WBS, schedule, cost baseline, risk register and stakeholder communications. If rejected or deferred, update the decision log and expectations.
10. Select corrective action for current variance. Test schedule compression only on the current critical path and state the new risk introduced.
11. Use the control-chart template to show whether pick accuracy has stabilised at or above the 98% success threshold. Distinguish control limits from specification.
12. Complete acceptance evidence for the three outlets, training, support handover, data ownership, operational KPIs, unresolved items and final vendor payment.
13. Open the retrospective template. Record what worked, what failed, the root cause, recommendation and named owner for organisational follow-up.
14. Issue the closure recommendation only when acceptance, transition, contracts, records, lessons learned and benefits ownership are complete.

Acceptance Criteria

Earned-value calculations and forecast reconcile; the change decision states complete impacts; corrective action has an owner and due date; and closure is supported by acceptance, handover, contract and lessons evidence.

Debrief

- Status reporting is a decision system: variance without forecast and action is only history.
- An approved change is not complete until every affected baseline and traceability record is updated.
- Project closure transfers accountability for benefits and operations; it does not merely stop project work.

Assessment Readiness

- Written Assessment (Case Study) — apply project management judgement to a small-project scenario.
- Oral Questioning — explain and justify decisions against the course methods and evidence.
- Open book: course slides, Learner Guide and approved course materials only.
- Answer structure: method, scenario evidence, result, trade-off and next action.

Glossary

Term	Definition
Acceptance criteria	Measurable conditions a deliverable must satisfy to be accepted.
Baseline	The approved scope, schedule or cost reference used to measure performance.
Business case	The justification comparing options, costs, benefits and risks before investment.
Charter	The document that authorises the project and grants the project manager authority.
Contingency reserve	Budget or time for identified risks whose responses are planned.
Corrective action	Action intended to bring future performance back in line with the plan.
Critical path	The longest-duration path through the schedule network, normally with least float.
Earned value	The budgeted value of work actually completed at the status date.
Float	Time an activity can move without affecting a defined schedule date.
Issue	A condition or event that has already occurred and requires action.
Milestone	A zero-duration marker for a significant event, approval or completion.
RACI	Responsible, Accountable, Consulted and Informed role mapping.
Residual risk	Risk remaining after the planned response is applied.
Risk trigger	Observable threshold or event indicating a risk is becoming imminent or has occurred.
Rolling-wave planning	Planning near-term work in detail while keeping later work at a higher level.
Scope creep	Uncontrolled addition of scope without approved time, cost or resource change.
WBS	Deliverable-oriented hierarchical decomposition of the total project scope.